

PROJECT REPORT

Artificial Intelligence Internship - Navitech Solution

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Project Title: Handwritten Digit Recognition using MNIST Dataset
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1. Introduction

This project explores the application of Deep Learning to recognize handwritten digits. Using the MNIST (Modified National Institute of Standards and Technology) dataset, we built and trained a neural network model. The significance of this project lies in its foundational approach to computer vision, which has far-reaching applications in pharmacy, such as automating prescription reading and microscopic botanical analysis.

2. Methodology (The 12-Step Process)

The implementation followed a structured 12-step engineering workflow:

Phase	Description
Data Setup	Importing libraries (TensorFlow, Keras, OpenCV) and loading the dataset.
Preprocessing	Scaling pixel values from [0,255] to [0,1] for mathematical stability.
Model Design	Building a Sequential Neural Network with Flatten and Dense layers.
Training	Optimizing the model using the Adam optimizer over 10 epochs.
Evaluation	Testing the model on unseen data and predicting sample digits.

3. Results & Visualization

The model demonstrated high efficiency during the training phase. Below is the visualization of a sample digit from the dataset used for training.

Sample Digit 3

Figure 1: Visualizing the 10th image (Digit 3) from the MNIST training set.

3.1 Model Performance

The model was trained for 10 epochs. The accuracy improved steadily, reaching approximately 98.9% on the training set and 97.08% on the test set.

Training Epochs

Figure 2: Execution of 10 Training Epochs showing Loss and Accuracy improvements.

4. Final Testing & Prediction

The final model was tested against the test dataset. In the verification step, the model correctly identified the handwritten digit '7', matching the ground truth label.

Prediction Results

Figure 3: Final Model Evaluation and Prediction of Digit '7'.

5. Conclusion & Pharmacy Application

The project successfully achieved a high accuracy of 97.08%. As a Master of Pharmacy student, I envision applying these deep learning principles to Pharmacognosy. Automated image recognition can be utilized to identify medicinal plants, detect adulterants in powdered drugs through microscopic image analysis, and digitize handwritten pharmaceutical records to reduce clinical errors.